

Feasibility of Organ Transportation by a Drone: An Experimental Study Using a Rat Model

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ABSTRACT

Background. Recently, the successful delivery of organs for transplantation using drones was reported. We investigated the influence of transportation by drones on the quality of liver grafts using a rat model.

Methods. Livers of 12 rats (8 and 32 weeks old) were divided into 2 groups of six. Livers were split into 2 parts and allocated to the drone or control groups (both $n = 12$). The drone experiment was conducted between islands in Nagasaki Prefecture, Japan. The distance between the islands was 12 km. Livers of the drone group were transported by a multicopter at a speed of 30 km/40 km/h over 60 m above sea level. Transported liver quality was analyzed by histology, and biochemistry data were compared between groups.

Results. Cold ischemia time did not differ between groups (902 min and 909 min, respectively). There were no differences in macroscopic findings regarding coloration and damage between groups. Aspartate aminotransferase (AST), alanine aminotransferase (ALT), and alkaline phosphatase (ALP) in preservation fluid were graft weight-corrected and compared, and no significant differences were found between groups: AST/g (4.61 vs 4.81 IU/L), ALT/g (2.78 vs 2.92 IU/L), and ALP/g (39.1 vs 37.0 IU/L). Immunochemical staining showed no significant difference between groups for terminal deoxynucleotidyl transferase dUTP nick and labeling staining (141 vs 113 cells), CD163 (818 vs 870 cells), and TNF- α (1.25 vs 1.41 scores).

Conclusions. The simulation experiment of organ transport for transplantation by drones was successfully conducted. There were no differences in the quality of livers transported by drones or other means. Further studies including large-animal experiments could lead to future clinical applications.

IN transplantation medicine, the graft ischemic time has a significant effect on outcomes [1–3]. Therefore, it is very important to plan for more appropriate logistics of organ transportation. In Japan, current organ transportation often involves chartered aircraft or helicopters for heart transplants, whereas other organs are often transported using public transport. Thus, the organ transportation system is based on a complex network comprising public transport, private transport, and medical workers in transplantation medicine [4].

If public transport is used, time adjustments are necessary because of the use of scheduled flights. In addition, there is a risk that traffic congestion will affect transportation. Therefore, the timing of organ transplants is often dependent on public transport.

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The accumulation of time resulting from this complex network can make transplantation impossible [5]. In recent years, pilot clinical studies of organ delivery by unmanned aerial systems were reported to solve these problems [6,7]. Unmanned aerial systems enable faster organ transplantation through smoother connections to aircraft and time saving by avoiding traffic jams [8,9]. They also improve access to transplantation in various situations, such as when major roads are closed to traffic due to accidents or disasters. The availability of unmanned aerial systems may also have the advantage of reducing the risk of accidents associated with the transport of organs for transplantation by transplant physicians and transplant-related staff.

There have been no reports of organ transportation by drones in Japan. In this study, we conducted an experiment to test the feasibility of transplant organ transportation by drones in Japan, using an animal model in cooperation with local governments and companies.

MATERIAL AND METHODS

Animal Model

Livers were obtained from model rats at the Nagasaki University Animal Research Facility (approval number: 211202-1). Twelve male Wistar rats (six 8-week-old and 6 32-week-old rats) were set up as models. Whole livers from model rats were obtained and University of Wisconsin (UW) solution was administered in 40 mL–50 mL flushes through the portal vein to perfuse the liver. The livers were split into 2 parts, weighed, grouped into drone and control groups, and then packed in accordance with the protocol of the Japanese Society for Transplantation (1st layer: 100 mL of UW solution, 2nd layer: 100 mL of saline solution, 3rd layer: no solution). Packed livers were stored in a cold container with ice. The above operations were performed on 12 rats per group. The rats were transported to the drone experiment site by a high-speed jetfoil and car.

Drone Transport Experiment

The experiment was conducted between Fukue Island and Hisaka Island, Goto City. The drone used was a multicopter type with a payload capacity of 5 kg and a maximum flight speed of 10 m/s (Fig 1). The livers in the drone group were transported by drone over a distance of approximately 12 km, whereas those in the control group were transported by high-speed



Fig 1. The six-winged multicopter drone used in this study.

vessel and car. Management and decision-making related to drones, such as drone operation and route determination, were carried out by a team of drone specialists. The criteria for the drone used in this study to be deemed flight-worthy was set at a maximum wind speed of 10 m/sec, and it flew along a predetermined route using autonomous navigation. The drone's position was constantly monitored using a Global Positioning System, and it was programmed to switch to manual control in case of any abnormalities. The transportation time and temperature inside the containers during transportation were recorded. After the transportation experiment between the islands was completed, specimens were transported to the Nagasaki University Experimental Facility by high-speed vessel and car, and the specimens were processed.

Biochemical Examination in Preservation Solution

The UW solution (packing layer 1) was collected after transport, stored at -30°C , and measured for aspartate aminotransferase (AST), alanine aminotransferase (ALT), and alkaline phosphatase (ALP) enzyme activities, which were evaluated as values per liver weight.

Histopathological Analyses

Liver tissues were processed for optical microscopy. For this, specimens were fixed in 4% formaldehyde for 48 hr–72 hr, encased in paraffin, and 5 μm -thick sections were prepared. Hematoxylin-eosin (HE) staining and immunohistochemical (IHC) staining were performed. Immunohistochemical staining was performed with Dako Target Retrieval Solution, Tris/EDTA solution (pH = 9.0), Dako REAL Peroxidase-Blocking Solution (Cat. No. S2023, Agilent Technologies, Santa Clara, CA), Protein Block Serum-Free Ready-to-use (Cat. No. X0909), Dako REAL Antibody Diluent (Cat. No. S2022), and Dako REAL EnVision/HRP secondary antibodies, Rabbit/Mouse (Cat. No. K5007). These procedures were performed in accordance with the protocol provided by Dako. After visualization with diaminobenzidine, the sections were counterstained with hematoxylin. The stained specimens were observed using an Olympus BX53 microscope (Olympus, Tokyo, Japan). The primary antibodies used were as follows: rabbit anti-CD163 antibody (Cat. ab182422, Abcam Inc., Cambridge, MA; 1:500) and anti-TNF- α antibody (Cat. ab220210, Abcam Inc.; 1:500).

Terminal deoxynucleotidyl transferase dUTP nick and labeling staining. Terminal deoxynucleotidyl transferase dUTP nick and labeling (TUNEL) staining was performed using a TUNEL Assay kit (Cat. ab206386, Abcam Inc., Cambridge, MA) in accordance with the manufacturer's instructions.

Histopathological and TUNEL staining evaluation. Hematoxylin-eosin staining was evaluated on the basis of the number of cases with findings of necrosis caused by external damage, balloon-like hepatocytes, and edema.

CD163 and TUNEL staining were evaluated quantitatively by counting the number of positive cells in 5 randomly selected fields of view ($\times 200$). TNF- α was evaluated semiquantitatively by scoring on a 4-point scale (0–3) as follows: 0: no expression, 1: mild overexpression, <25% of hepatocytes, 2: moderate overexpression, 25%–50% of hepatocytes, and 3: high overexpression, >50% of hepatocytes.

Statistical Analysis

The data for continuous variables were expressed as means with standard deviation. Data of biochemical, histopathological, and TUNEL staining data were analyzed by two-tailed *t* test using JMP Pro (SAS Institute Inc., Cary, NC). A *P* value <.05 was considered statistically significant.

Table 1. Liver Weight and Cold Ischemia Time in Each Group

Age in weeks	Drone		Control		P value
	Young	Old	Young	Old	
Number of samples	6	6	6	6	-
Liver weight(g)	4.3±0.2	6.9±0.5	9.3±0.4	14.8±2.3	<.01
CIT(min)	895±32	909±26	903±34	915±29	.57

RESULTS

Liver Weight and Cold Ischemic Time

The drone and control groups each consisted of 6 young rat livers and 6 old rat livers. The lighter weight split livers were assigned to the drone group. The mean liver weight was 4.3 ± 0.17 g for young rats and 6.9 ± 0.45 g for old rats in the drone group, and 9.3 ± 0.4 g for young rats and 14.8 ± 2.3 g for old rats in the control group. The mean cold ischemic time was 894 ± 32 min for young rat livers and 909 ± 26 min for old rat livers in the drone group, and 903 ± 34 min for young rat livers and 915 ± 29 min for old rat livers in the control group (Table 1). There was no significant difference in cold ischemia time between the drone and control groups ($P = .57$).

Temperature inside the container during transport. The outside temperature during the experiment was 12°C - 16°C . The temperature inside the container in the drone group was maintained at 0°C - 5°C and did not vary depending on the method of transportation (Fig 2).

Biochemical examination of preserved solution samples. The values of AST, ALT, and ALP (per liver weight) in the stored solution of the drone and control groups were AST/g: 4.61 ± 3.06 vs 4.84 ± 1.52 ($P = .81$), ALT/g: 2.78 ± 2.32 vs 2.92 ± 1.04 ($P = .85$), and LDH/g: 39.1 ± 31.1 vs 37.0 ± 7.6 ($P = .82$), respectively (Table 2). There were no significant differences in the AST/g, ALT/g, and ALP/g in the preserved solution between the 2 groups.

Macroscopic findings. Macroscopic findings of the livers after transportation are shown in Fig 3. There were no differences in the macroscopic findings of livers from both groups based on necrosis, color change, or external damage.

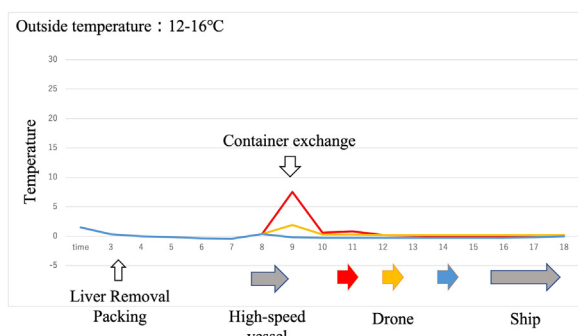


Fig 2. Temperature inside the container. The temperature inside the containers remained constant regardless of the transfer method.

Table 2. Biochemical Analysis

	Drone	Control	P value
AST/g	4.6±3.0	4.8±1.5	.81
ALT/g	2.8±2.3	2.9±1.0	.85
LDH/g	39.1±31.1	37.0±7.6	.82

HE staining. The results of HE staining are shown in Fig 4. There were no findings of necrosis caused by external damage, ballooning hepatocytes, or edema in any of the livers, and no findings suggestive of effects from drone transportation.

TUNEL staining and immunochemical staining. The results of TUNEL staining and immunochemical staining are shown in Fig 5. TUNEL staining showed that the mean quantitative values of the drone and control groups were 141 ± 274 vs 113 ± 211 cells ($P = .78$), and there was no significant difference between the 2 groups in the assessment of apoptosis. There were no significant differences in the number of Kupffer cells (818 ± 92 vs 870 ± 77 cells; $P = .14$) or inflammatory markers (CD163 and TNF- α) (1.25 ± 0.45 vs 1.41 ± 0.51 scores; $P = .4$).

DISCUSSION

An organ transportation experiment was conducted using a drone to transport rat livers. Temperatures were maintained at a constant level during transport, and no adverse effects of drone transportation were observed regarding the physiological or pathologic evaluation of the specimens. There was no significant effect on livers, confirming the safety of organ transportation by drones.

Regarding the current organ transportation system, there are various time issues associated with the use of public transport.

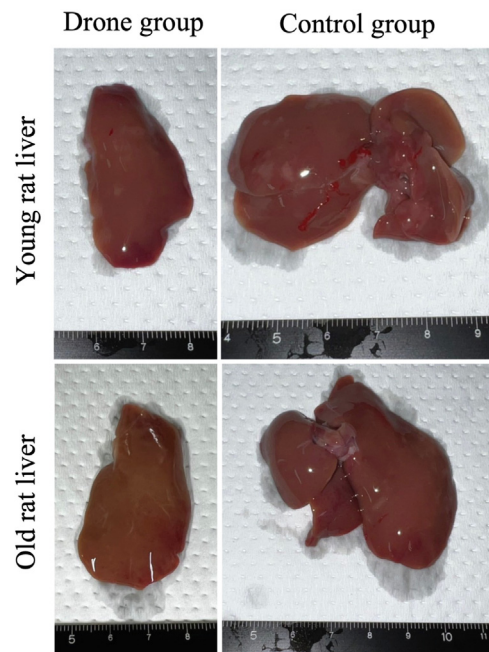


Fig 3. Macroscopic analysis of livers. There were no obvious findings of necrosis, color change, or external damage.

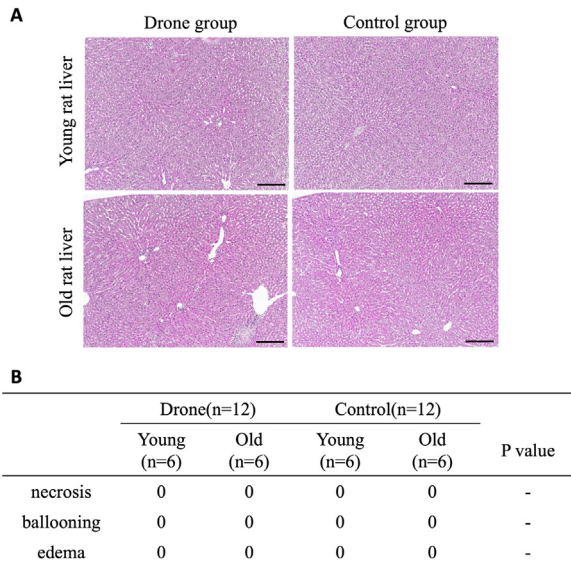


Fig 4. Hematoxylin-eosin (HE) staining. There were no cases of necrosis caused by external influences, balloon-like hepatocytes, or edema. Scale bar, 200 μ m.

For example, staff transporting organs may have to wait because of traffic congestion or lose time because of limited numbers of flights. If drones can be used as a means of organ transportation, they can avoid traffic congestion and connections to scheduled flights in the shortest possible time by taking over the responsibility of transporting the organs from the hospital to the airport. As a result, the cold ischemia time could be shortened. In addition, if future improvements in drone performance enable long-distance, stable flights and direct inter-hospital transport, organ transport will be possible without relying on aircraft time, and further cold ischemia time reduction can be expected.

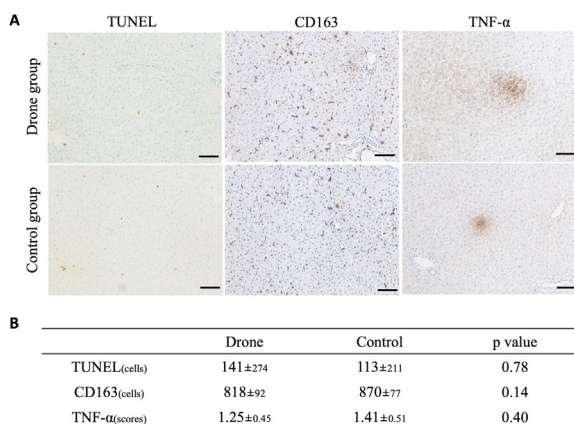


Fig 5. Terminal deoxynucleotidyl transferase dUTP nick and labeling (TUNEL) staining and immunochemical analyses. TUNEL staining and immunochemical staining in young rat livers are shown. No significant differences in staining were found between the drone and control groups. Scale bars, 100 μ m.

Transplant medicine is considered a high-risk procedure in medicine. In 2007, 6 members of the University of Michigan organ transplantation team died during lung transportation [10]. In Japan in 2020, an organ removal team from the University of Tokyo crash-landed while transporting a heart by helicopter, resulting in serious injuries [11]. Thus, the current organ transportation system puts transplantation teams at risk of being involved in accidents. We think that unmanned drones can perform part of this transportation, thereby reducing the risk of accidents to the transplantation team.

In December 2022, the Japanese Civil Aeronautics Law was revised to allow unobserved flights in manned areas if certain conditions are met [12]. This may lead to increased opportunities for drones to be used in the future for the transportation of pharmaceuticals and medical materials, in addition to the delivery of food and other goods. This study provides basic data for considering the feasibility of organ transportation using drones in future transplantation medicine.

Regarding the temperature management of organs during transportation, currently drones do not have temperature control capabilities. Therefore, in this study, we placed a temperature logger inside the container storing the organs and checked the temperature with the logger after landing. Currently, even with conventional means of organ transportation, continuous temperature monitoring is not being conducted. Therefore, if continuous temperature monitoring becomes necessary for organ transportation by drones in the future, the development of a temperature management system would also be required.

The drone used in this study had a maximum payload capacity of 5 kg and was tested using livers from rats. A drone that can withstand a weight of about 10 kg is needed to transport grafted human livers in actual clinical practice. In addition, this experiment involved air transport by a drone over a distance of approximately 12 km. The longest distance achieved in recent drone-based organ transport experiments is approximately 14 km [6,7,13,14]. The safety of long-distance transportation using drones is still not fully understood. In terms of drone logistics, initially focusing on short-distance usage is desirable. Specifically, using drones for transportation between hospitals in urban areas or between hospitals and airports is a viable approach. This allows for the avoidance of risks associated with traffic congestion, disasters, and issues on roads, while enabling smooth connections to limited scheduled flights or charter aircraft. With the future improvement of drone performance and establishment of further safety measures in drone transport, it is anticipated that direct transportation between distant hospitals will become possible. This opens up opportunities for additional research in this field.

In conclusion, this study showed the feasibility of rat organ transport by a drone. Further simulation of organ delivery using larger organs may lead to the clinical use of drones in the future.

DATA AVAILABILITY

All data generated or analyzed during this study are included in this article. Further inquiries can be directed to the corresponding author.

DECLARATION OF COMPETING INTEREST

All the authors declare no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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